

Homework 3

Name:

Class:

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To remember how the matrix of T is constructed from T , you might write across the top of the matrix the basis vectors $v_1, \dots, v_n$ for the domain and along the left the basis vectors $w_1, \dots, w_m$ for the vector space into which T maps, as follows:

$$\begin{array}{c} w_1 \\ \vdots \\ w_m \end{array} \begin{pmatrix} & v_1 & \cdots & v_k & \cdots & v_n \\ & & & A_{1,k} & & \\ & & & \vdots & & \\ & & & A_{m,k} & & \end{pmatrix}$$

The k th column of the matrix representation of T consists of the scalars needed to write Tv_k as a linear combination of $w_1, \dots, w_m$:

$$Tv_k = \sum_{j=1}^m A_{j,k} w_j$$

1. Let $M_{n \times n}$ be the vector space of all $n \times n$ matrices, and define $T : M_{n \times n} \rightarrow M_{n \times n}$ by $T(A) = A + A^T$.

(a) Show that T is a linear transformation.

(b) If $n = 2$:

(i) Show that the range of T is the set of B in $M_{2 \times 2}$ with the property that $B^T = B$.

(ii) Describe the null space of T .

(iii) Find a basis of range T and null T and state the dimension of them.

(c) Solve (b)(iii) again when n is an arbitrary positive integer ($n \geq 2$).

(d) 【选做】 Describe the relationship between $M_{n \times n}$, range T and null T .

2. Suppose $\mathbf{v}_1, \dots, \mathbf{v}_m$ is a list of vectors in V . Define $T \in \mathcal{L}(\mathbb{R}^m, V)$ by

$$T(\mathbf{x}) = x_1 \mathbf{v}_1 + \cdots + x_m \mathbf{v}_m,$$

$$\text{for } \mathbf{x} = \begin{bmatrix} x_1 & \cdots & x_m \end{bmatrix}^T \in \mathbb{R}^m.$$

- (a) What property of T corresponds to $\mathbf{v}_1, \dots, \mathbf{v}_m$ spanning V ? (Injective or Surjective.) Why?
- (b) What property of T corresponds to $\mathbf{v}_1, \dots, \mathbf{v}_m$ being linearly independent? (Injective or Surjective.) Why?

3. Let $E = \{u_1, u_2, u_3\}$ and $F = \{b_1, b_2\}$, where

$$u_1 = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}, \quad u_2 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \quad u_3 = \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$$

and

$$b_1 = (1, -1)^T, \quad b_2 = (2, -1)^T.$$

For each of the following linear transformations L from $\mathbb{R}^3$ into $\mathbb{R}^2$, find the matrix representing L with respect to the ordered bases E and F :

$$(i) L(x) = \begin{pmatrix} x_1 + x_2 \\ x_1 - x_3 \end{pmatrix}; \quad (ii) L(x) = \begin{pmatrix} 2x_2 \\ -x_1 \end{pmatrix}.$$

4. Find the standard matrices A of the following linear transformations.

- (a) $T \in \mathcal{L}(\mathbb{R}^2, \mathbb{R}^2)$ is a vertical shear transformation that leaves $\mathbf{e}_2$ unchanged and maps $\mathbf{e}_1$ into $2\mathbf{e}_2 + \mathbf{e}_1$.
- (b) $T \in \mathcal{L}(\mathbb{R}^2, \mathbb{R}^2)$ first performs a horizontal shear transformation that leaves $\mathbf{e}_1$ unchanged and maps $\mathbf{e}_2$ into $-2\mathbf{e}_1 + \mathbf{e}_2$ and then reflects points through the line $x_2 = x_1$.

5. Let V be the subspace of $C[a, b]$ (continuous functions on $[a, b]$) spanned by $1, e^x, e^{-x}$, and let D be the differentiation operator on V .

- (a) Find the matrix A representing D with respect to the ordered basis $[1, \cosh x, \sinh x]$. [$\cosh x = \frac{1}{2}(e^x + e^{-x})$, $\sinh x = \frac{1}{2}(e^x - e^{-x})$.]
- (b) Find the matrix B representing D with respect to $[1, e^x, e^{-x}]$.
- (c) Find the invertible matrix S such that $B = S^{-1}AS$. (Hint: transition matrix)

6. 【选做：几何拓展。不计入成绩】The *homogeneous* coordinate system is formed by equating each vector in $\mathbb{R}^2$ with a vector in $\mathbb{R}^3$ having the same first two coordinates and having 1 as its third coordinate.

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \leftrightarrow \begin{bmatrix} x_1 \\ x_2 \\ 1 \end{bmatrix}$$

When we want to plot a point represented by the homogeneous coordinate vector $(x_1, x_2, 1)^T$, we simply ignore the third coordinate and plot the ordered pair (x_1, x_2) . Let

$$R = \begin{bmatrix} 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

The column vectors of R represent the homogeneous coordinates of points in the plane.

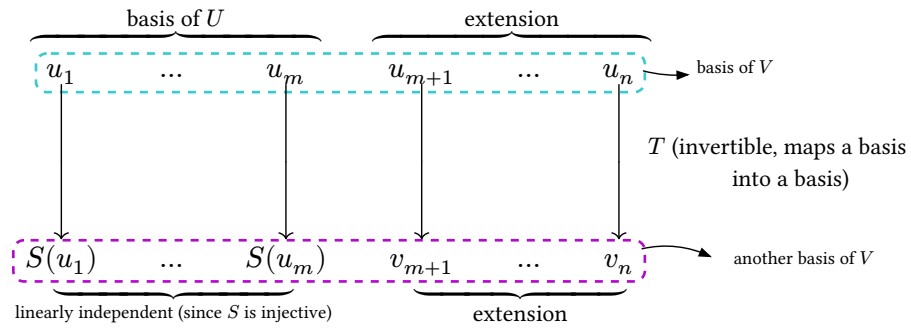
- Draw the figure whose vertices correspond to the column vectors of R . What type of figure is it?
- For each of the following choices of A , sketch the graph of the figure represented by AR and observe the effect of the linear transformation. [Hint: pay special attention to (c). Think: why homogeneous coordinate?]

$$(1) A = \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{3} & 0 \\ 0 & 0 & 1 \end{bmatrix}; \quad (2) A = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ \frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}; \quad (3) A = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & -2 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 3 \\ 0 & 0 & 1 \end{bmatrix}.$$

7. (a) Suppose $T \in \mathcal{L}(V, W)$ is injective and $v_1, \dots, v_n$ is linearly independent in V . Prove that $T(v_1), \dots, T(v_n)$ is linearly independent in W .
- (b) Suppose V is finite-dimensional and that $T \in \mathcal{L}(V, W)$. Prove that there exists a subspace U of V such that $U \cap \text{null } T = \{0\}$ and $T(V) = T(U)$. Find a basis.

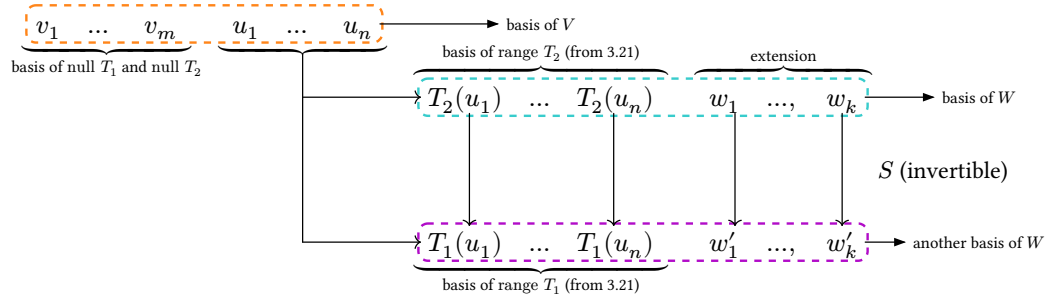
8. (a) Suppose V is finite-dimensional, U is a subspace of V , and $S \in \mathcal{L}(U, V)$. Prove there exists an invertible operator $T \in \mathcal{L}(V)$ such that $T(u) = S(u)$ for every $u \in U$ if and only if S is injective.

Hint for (a):



- (b) Suppose V, W are finite-dimensional and $T_1, T_2 \in \mathcal{L}(V, W)$. Prove that $\text{null } T_1 = \text{null } T_2$ if and only if there exists an invertible operator $S \in \mathcal{L}(W)$ such that $T_1 = ST_2$.

Hint for (b):



9. Suppose V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that $\dim \text{range } T = 1$ if and only if there exist a basis of V and a basis of W such that with respect to these bases, all entries of the matrix representation $\mathcal{M}(T)$ equal 1.

10. Suppose that V is a vector space. Show that V and $\mathcal{L}(\mathbb{F}, V)$ are isomorphic vector spaces. (**Note:** V is not necessarily finite-dimensional, so you can't utilize the dimension test, and you should try to construct an isomorphic linear map.)